Juan Contreras

Software Engineer - Games (Unreal Engine 5 / Unity / C++ / C#)

Detroit, MI | juan@codejuan.com | codejuan.com/games | github.com/Code-Juan | linkedin.com/in/juan-contreras68

// SUMMARY

Early-career game engineer who builds gameplay, AI, animation, and state-management systems in Unreal Engine 5 and Unity (C++ and C#) and works from design documents and acceptance criteria on cross-discipline teams. B.S. in Game Development (Computer Science) - valedictorian, 3.76 GPA - plus professional software experience building and maintaining production systems. Strong data structures and algorithms foundation.

// TECHNICAL SKILLS

Engines: Unreal Engine 5 (C++, Blueprints, animation blueprints, component architecture, child actors), Unity (C#, NavMeshAgent AI, Cinemachine, ScriptableObjects, async scene loading, save/load), Vulkan API with Gateware framework (custom-engine C++ gameplay)

Languages: C++, C#, Python, JavaScript/TypeScript, SQL

Algorithms & Data Structures: Binary search trees, hash tables, linked lists, dynamic arrays, Huffman coding, recursion, Shunting Yard parsing, RAII and memory management, performance-minded implementation

Practices: Agile/Scrum sprints, Jira epics with story points and acceptance criteria, documented Git branching strategies, unit testing (50+ case suites), code reviews, systematic debugging, design documentation, cross-discipline collaboration

// GAME PROJECTS

Nutty Pursuit - C++ arcade game on Vulkan (5-person team, playable) — Oct 2025

- Built player combat and control systems - 8-directional shooting with normalized velocity - for a 3-level, 2.5D arcade game; worked from Jira epics with story points and acceptance criteria under a documented Git branching strategy that kept every sprint release demo-ready.

Fluxuate - Unreal Engine 5 team game (in development) — Dec 2025 - Feb 2026

- Engine/UI programmer: built the 3-slot save system, game-state flow, and menu/UI framework, plus climbing (component reparenting), weapon animation, movement, and health/stamina systems from a shared design document; diagnosed and fixed an enemy-persistence save bug across successive builds.

Earth Untamed - Unity 6 vertical slice (5-person capstone team) — Jan - Feb 2026

- Built an interface-driven weapon system on ScriptableObject data with editor tooling, animation-event hit detection so swings land when they visually connect, a player audio system synced to UI state, and climbing and viewmodel overhauls.

Project Galaxy - Unity 3D FPS in C# (5-person Scrum team) — Dec 2024 - Jan 2025

- Built an enemy AI hierarchy on an abstract NavMeshAgent base class (including a stun enemy that steals from the player's inventory) and an ability-based boss, a ScriptableObject inventory behind a singleton manager, JSON save/load with GUID scene persistence, and async scene loading through Alpha, Beta, and QA sprints.

END2508 - Unreal Engine 5 C++ systems — Aug 2025

- Component-based character architecture: inheritance-based base classes, health component, child-actor weapon system, event-driven design, animation blueprint integration with attention to memory management.

Algorithms & tools - C++ — 2024

- Expression calculator (Singleton and Factory Method patterns, Shunting Yard parsing, 50+ unit tests); Conway's Game of Life (resolved 15+ rendering and input defects through systematic debugging); data-structures implementations (BSTs, hash tables, Huffman coding).

// PROFESSIONAL EXPERIENCE

Web Development & IT Manager - The Wooden Stone, LLC (Clinton Township, MI) — March 2025 - Present

- Build and maintain production software end to end - the company's internal operations platform (Next.js, TypeScript, Postgres - including its estimating engine, checked by 1,400+ automated tests), two customer-facing web services (Node.js/Express, REST integrations), automation that cut manual data entry by 30%, and root-cause fixes documented for a non-technical team - managing workload and task status independently while completing a second degree as valedictorian.

// EDUCATION

B.S., Game Development (Computer Science) - Full Sail University - Valedictorian, 3.76 GPA — March 2026

B.S., Industrial & Systems Engineering - University of Michigan-Dearborn — April 2023